

7955 – ECONOMETRICS

Report

Brent Oil Prices

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1 Introduction

Crude oil is central to transportation, industry and power generation, so daily oil moves transmit quickly to inflation, corporate costs and financial markets. Brent is the leading global benchmark. At the daily frequency, prices are nonstationary, returns are volatile, heavy-tailed, and only weakly predictable in the mean.

This report asks: (i) does yesterday's return help explain today's Brent return? (ii) is this relationship different across the distribution (e.g., in the tails)? I work with daily log-returns (in percent) and estimate: a baseline OLS of r_t on r_{t-1} ; an extended OLS adding r_{t-2} and $|r_{t-1}|$ with a joint F-test; and quantile regressions across $\tau \in \{0.1, \dots, 0.9\}$.

Preview of results. Baseline OLS shows a very small slope and $R^2 \simeq 0.00058$. The extended model lifts R^2 to about 0.00134; the joint F-test on the added regressors rejects at 5% ($F = 3.16$, $p = 0.042$), but the economic gain is modest. Quantile regressions indicate mild heterogeneity with wider uncertainty in the tails (positive slope in lower quantiles, negative in the upper tail).

2 Data

I use the Kaggle *Brent Oil Prices* daily spot series (USD/barrel). I harmonize the date field, sort chronologically, and compute daily log-returns $r_t = 100 [\log(P_t) - \log(P_{t-1})]$. Sample span: 1987-05-25 to 2020-04-21. After removing the initial NA and merging lags, the cleaned sample used in the regressions has **8,358** daily returns.

Key descriptives (returns r_t). Clean sample size: **8,357** days (1987-05-25 to 2020-04-21).

$$\begin{aligned} \min &= -64.37\%, & Q_1 &= -1.115\%, & \text{Median} &\approx 0.040\%, \\ \text{Mean} &= 0.0179\%, & Q_3 &= 1.213\%, & \max &= 41.20\%, & \text{sd} &= 2.553\%. \end{aligned}$$

	Mean (%)	Var	SD	Skew	Kurtosis	Range (%)
r_t	0.0179	6.5189	2.5532	-1.744	65.852	[-64.37, 41.20]

	Mean (%)	SD	Q_1 (%)	Median (%)	Q_3 (%)	Range (%)
r_t	0.0100	2.4962	-1.1156	0.0396	1.2110	[-64.37, 22.16]

Figures

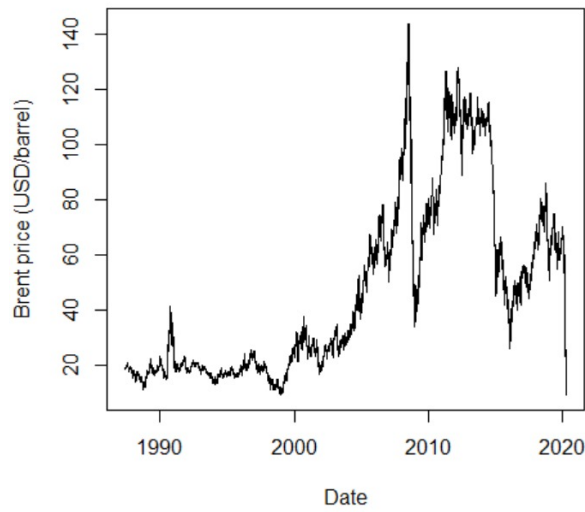


Figure 1: Daily Brent price (USD/barrel).

Note. Large swings and episodes (2008, mid-2010s, 2020) justify modelling returns rather than levels.

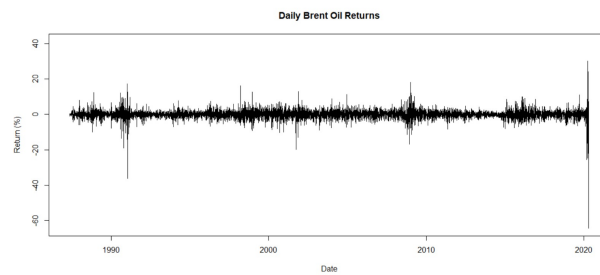


Figure 2: Daily Brent returns (log, %).

Note. Centered near zero, with volatility clustering.

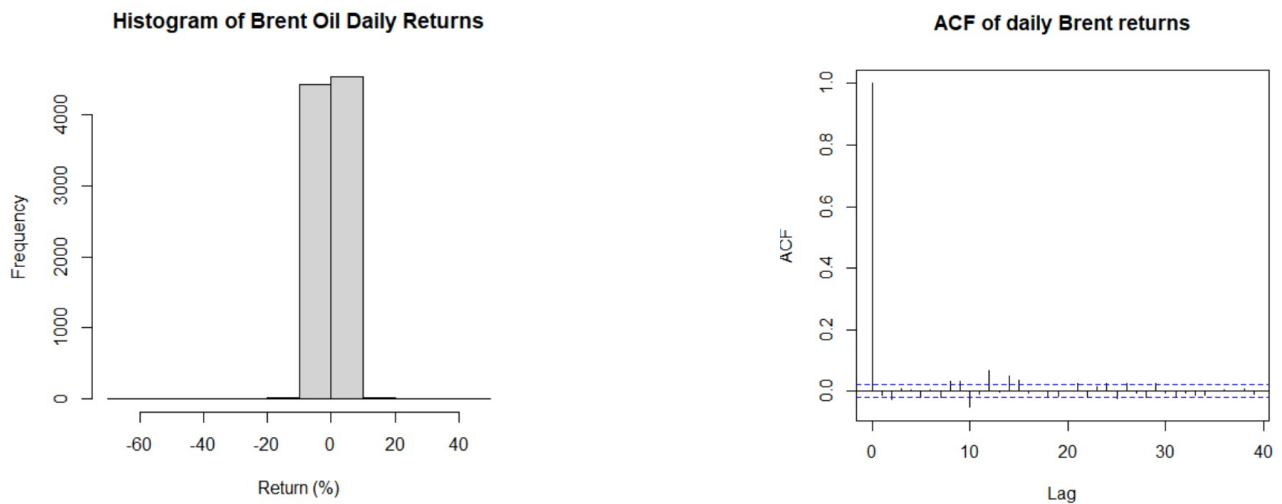


Figure 3: Histogram of returns (left); ACF of returns (right).

Note. Mass near zero with pronounced tails; ACF shows near-zero linear autocorrelations at daily frequency.

3 Methodology

Models and identification

Baseline conditional mean:

$$r_t = \alpha + \beta_1 r_{t-1} + u_t.$$

Extended specification:

$$r_t = \alpha + \beta_1 r_{t-1} + \beta_2 r_{t-2} + \beta_3 |r_{t-1}| + u_t.$$

I test $H_0 : \beta_2 = \beta_3 = 0$ using an extra sum-of-squares (joint) F-test. To assess distributional heterogeneity, I estimate

$$Q_\tau(r_t | r_{t-1}) = \alpha_\tau + \beta_{1,\tau} r_{t-1}, \quad \tau \in \{0.1, \dots, 0.9\},$$

and report 95% confidence bands.

Why this design

Daily oil prices are nonstationary and exhibit strong long-run trends. Using returns instead of prices removes these trends and produces a stationary series, which is a prerequisite for valid OLS inference (Chapter 2). The first lag r_{t-1} is included to capture very short-run predictability such as momentum (positive persistence) or mean reversion (negative persistence), both of which are common hypotheses in financial markets (Chapter 3). Adding the second lag r_{t-2} tests whether any linear dependence extends beyond one day, in line with the multiple regression framework.

The term $|r_{t-1}|$ is motivated by volatility clustering, a well-documented feature of financial time series: large shocks tend to be followed by large shocks, even if the conditional mean remains close to zero. Including $|r_{t-1}|$ provides a simple proxy for conditional variance without estimating a full GARCH model (Chapter 5).

Finally, OLS focuses only on the conditional mean, which may hide heterogeneous effects across the distribution. Quantile regression allows us to evaluate whether the impact of r_{t-1} is stronger in market stress episodes (lower tail), in rallies (upper tail), or mainly around the center of the distribution. This distributional perspective is central in Chapter 6 and provides a richer view of short-run dynamics than mean regression alone.

Origin of the formulas used

All formulas applied in this report come directly from the course:

- **OLS model** $r_t = \alpha + \beta_1 r_{t-1} + u_t$ comes from *Chapter 2: The Classical Linear Regression Model*. The formulas for OLS estimators, t -statistics, confidence intervals, and R^2 are the ones introduced in that chapter.
- **Multiple regression** $r_t = \alpha + \beta_1 r_{t-1} + \beta_2 r_{t-2} + \beta_3 |r_{t-1}| + u_t$ relies on *Chapter 3: The Multiple Regression Model*, including interpretation of coefficients, joint significance, and adjusted R^2 .
- **Joint F-test** $H_0 : \beta_2 = \beta_3 = 0$ follows the extra-sum-of-squares methodology of *Chapter 3*, where restricted vs unrestricted models are compared.
- **Diagnostic tests** such as Breusch-Pagan are taken from *Chapter 6: Classical Linear Regression Model Assumptions And Diagnostic Tests*, including heteroskedasticity tests and residual analysis.
- **Quantile regression** comes from *Chapter 4: Quantile Regression*, including the model $Q_\tau(r_t | r_{t-1}) = \alpha_\tau + \beta_{1,\tau} r_{t-1}$ and confidence bands.

4 Results

Descriptive evidence

Returns are centred near zero with heavy tails and volatility clustering. The ACF of returns shows only tiny spikes around zero, pointing to limited linear predictability of the conditional mean at a daily horizon.

Baseline OLS (Q1: weak mean predictability)

Estimating $r_t = \alpha + \beta_1 r_{t-1} + u_t$ gives (my run):

$$\hat{\alpha} = -0.008478 \text{ (SE } 0.027208, t = -0.3116, p = 0.7554), \quad \hat{\beta}_1 = 0.025166 \text{ (SE } 0.011403, t = 2.2069, p = 0.02735).$$

$R^2 = 0.0005826$ and $\text{adj. } R^2 = 0.000463$. The intercept is negligible; the slope is small although statistically different from zero at 5%. Economically the effect is tiny: a 1 pp move yesterday is associated with only ≈ 0.025 pp today on average.

Extended OLS (small statistical gain, limited economic value)

Adding r_{t-2} and $|r_{t-1}|$ yields:

$$\hat{\alpha} = 0.05149 \text{ (SE } 0.03695, p = 0.1635), \quad \hat{\beta}_1 = 0.02304 \text{ (SE } 0.01144, p = 0.0440), \quad \hat{\beta}_2 = -0.00994 \text{ (SE } 0.01143, p = 0.3846),$$

$R^2 = 0.001339$ ($\text{adj. } R^2 = 0.000981$), i.e., $\Delta R^2 \simeq 0.000756$ vs M_1 . A joint F-test rejects $H_0 : \beta_2 = \beta_3 = 0$ at 5% ($F = 3.1644, p = 0.04229$). The gain is statistically detectable but economically minor: fitted means stay close to zero and miss extremes.

Model evaluation and goodness of fit

Beyond coefficient significance, it is informative to evaluate the overall fit of the mean equation. In both specifications, the R^2 remains extremely small, which is typical for daily financial returns whose conditional mean is notoriously hard to predict. This is fully consistent with the econometric intuition developed in class: unless strong serial dependence exists, linear models explain almost none of the day-to-day return variation.

The joint F-test in the extended model confirms that adding r_{t-2} and $|r_{t-1}|$ improves the model statistically, yet the magnitude of the improvement is economically negligible. Most of the dynamics appear to reside not in the conditional mean but in the conditional variance, as suggested by the significant coefficient on $|r_{t-1}|$. This echoes a classical finding in financial econometrics: volatility, not the mean, drives short-run return behavior.

Distributional heterogeneity (quantile regression)

Quantile slopes vary mildly with wider uncertainty in the tails:

$$\beta_{1,0.10} = 0.05475 \text{ (} p = 0.00958), \quad \beta_{1,0.20} = 0.04776 \text{ (} p = 0.00047), \quad \beta_{1,0.90} = -0.04901 \text{ (} p = 0.01102).$$

This suggests modest asymmetries: slightly positive association in lower quantiles (selloffs) and negative at the top tail.

Comparison: OLS versus quantile regression

A key insight from this project is the contrast between OLS and quantile regression. OLS estimates the conditional mean of returns and therefore summarizes the central tendency only. In contrast, quantile regression provides a full view of how past returns affect different parts of the conditional distribution. This is particularly relevant for financial data, which are heavy-tailed and often asymmetric.

In this dataset, quantile regression reveals patterns that OLS cannot detect: mildly positive dependence in the lower quantiles and negative dependence in the upper tail. These effects are modest but statistically meaningful and highlight that dependence can vary across market conditions (e.g., selloffs versus strong rallies). Such distributional asymmetries are economically relevant, especially for risk management and tail-event analysis.

Diagnostics

Residual ACF is innocuous. A Breusch–Pagan test detects heteroskedasticity ($LM = 47.50$, $p = 5.50 \times 10^{-12}$), consistent with volatility clustering and the very low R^2 in mean equations.

To complement the Breusch–Pagan test, I examine the residual distribution. A Jarque–Bera test strongly rejects normality, reflecting heavy tails and asymmetry in the residuals, which is fully expected for high-frequency financial data. The residuals-versus-fitted plot shows a clear increase in dispersion for larger fitted values, reinforcing the presence of heteroskedasticity.

These diagnostics matter because the classical OLS inference framework relies on homoscedastic and approximately normal errors. While OLS point estimates remain unbiased, the standard errors become unreliable under heteroskedasticity unless robust methods (e.g., White standard errors) are used. The patterns observed here further motivate variance-based models, such as GARCH, when the goal is to capture the dynamic structure of returns.

Compact result tables

Model	$\hat{\alpha}$	SE(α)	$p(\alpha)$	$\hat{\beta}_1$	SE(β_1)	$p(\beta_1)$
$M_1 : r_t \sim r_{t-1}$	-0.00848	0.02721	0.7554	0.02517	0.01140	0.02735

M_2 term	$\hat{\beta}$	SE	t	p	R^2
r_{t-1}	0.02304	0.01144	2.014	0.0440	
r_{t-2}	-0.00994	0.01143	-0.869	0.3846	0.001339
$ r_{t-1} $	-0.03722	0.01552	-2.398	0.0165	

$\Delta R^2 \approx 0.0009806$; Joint F-test on (β_2, β_3) : $F = 3.1644$, $p = 0.04229$.

Figures

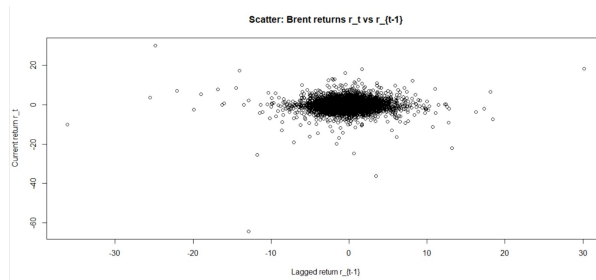


Figure 4: r_t vs r_{t-1} : dense cloud, weak linear association, some outliers.

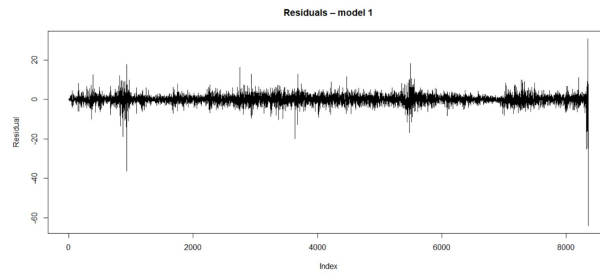


Figure 5: Residuals – model 1: centered near zero; clustering persists.

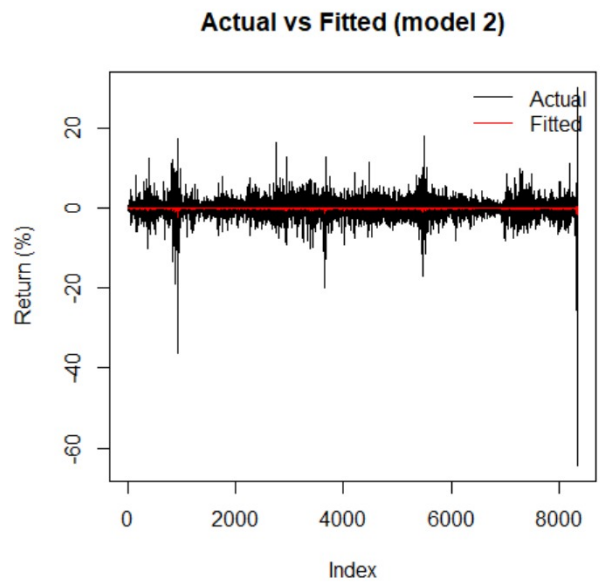


Figure 6: Actual vs fitted (extended OLS): fitted mean near zero; extremes missed.

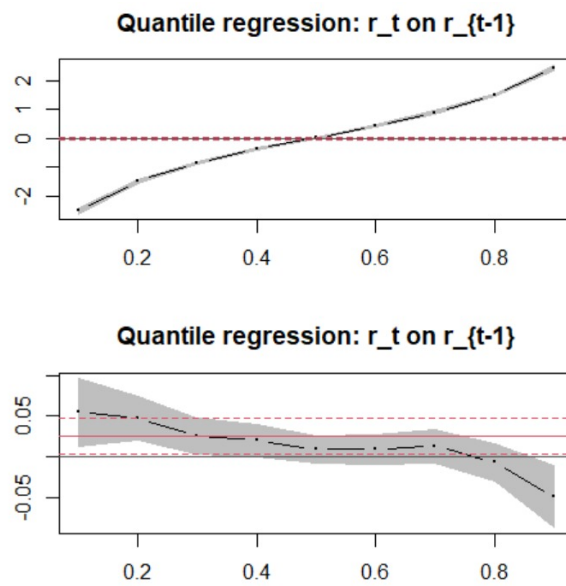


Figure 7: Quantile regression profile: modest slope variation; wider bands in tails (95% CI).

5 Discussion and financial interpretation

Overall, the results show that daily Brent oil returns behave like most liquid financial assets. The conditional mean is almost unpredictable, which explains why the coefficients in the OLS regressions are very small and why the R^2 values remain close to zero. This is consistent with the idea that commodity markets are close to informational efficiency at the daily frequency.

The data also display clear signs of heteroskedasticity and heavy tails. This means that the size of returns changes over time and that extreme movements are more frequent than under a normal distribution. The significant coefficient on $|r_{t-1}|$ reinforces this idea: large shocks tend to be followed by more volatile periods.

Quantile regressions provide a more detailed picture. They show slightly positive dependence in the lower quantiles (during market drops) and negative dependence in the upper tail (after strong price jumps). These effects are small, but they suggest that market dynamics can differ between stressful periods and more optimistic phases.

From a financial point of view, these findings confirm that most of the short-run dynamics in oil markets come from volatility rather than from the mean. Simple autoregressive models provide very limited predictive power for returns, while volatility behaves in a more structured way. This naturally points to models focusing on variance dynamics, such as GARCH, or to the use of external factors (e.g., equity markets, USD, macro conditions) to better explain and forecast movements in oil prices.

6 Conclusion

Q1 (predictability). The results show that yesterday's return has almost no ability to predict today's Brent return. The estimated slope is very small ($\hat{\beta}_1 \approx 0.025$) and the R^2 is close to zero, which confirms the weak predictability of daily oil returns.

Q2 (distributional differences). Quantile regressions reveal small differences across the distribution: the slope is slightly positive in the lower quantiles and slightly negative in the upper quantiles. However, these effects remain limited and do not change the overall conclusion that mean predictability is extremely weak.

Adding r_{t-2} and $|r_{t-1}|$ improves the model statistically (joint F-test $p = 0.04229$), but the economic impact remains very small ($R^2 \approx 0.00134$). The diagnostics clearly indicate strong heteroskedasticity, meaning that most of the dynamics come from changes in volatility rather than from the conditional mean.

Limitations and next steps. A simple linear mean model cannot capture volatility clustering or time-varying uncertainty. More advanced approaches such as GARCH models, the inclusion of exogenous variables (e.g. USD, VIX, equity indices), or analysis by subperiods (2008 crisis, 2014–2016 oil collapse, 2020 COVID shock) could provide a richer understanding of Brent price behaviour.

References

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- [5] Kaggle (n.d.). *Brent Oil Prices Dataset*. <https://www.kaggle.com/datasets/mabusalah/brent-oil-prices/data> (Data source)